

Martin, H. N. The human body; an account of its structure and activities and the conditions of its healthy working. 8th ed. 1904. 22cm
(Amer. sci. ser. Adv.) Holt \$2.50
Standard college text-book in human anatomy and physiology. Ed. sel.

Peabody, J. E. Studies in physiology, anatomy and hygiene. 1903. 19½cm
Macmillan \$1.10 n

613 Personal hygiene

Baker, M. N. Potable water and methods of detecting impurities. 1899. 15cm (Van Nostrand sci. ser.)
Van Nostrand 50c

Object to present clearly and briefly essential qualities of potable water, how it may be obtained, and significance of chemical, bacterial and microscopic tests. Discusses value of pure water, relations between water and disease and typhoid fever records as index of purity or impurity of public water supplies. *Engin.* n. 42: 278.

Bergey, D. H. Principles of hygiene. 1901. 24cm
Saunders \$3 n

Accurate; useful to medical students, physicians and health officers; also to architects in giving sanitary requirements for buildings. N. Y.

Billings, J. S. ed. Physiological aspects of the liquor problem; investigations made by and under direction of sub-committee of the Committee of fifty to investigate the liquor problem. 2 v. 1903. 23cm
Houghton \$4.50 n

Holds that use of alcohol impairs mental work, lessens physical power, does not protect against cold and fatigue, and diminishes resistance to infectious diseases; condemns and exposes fallacy of some misdirected efforts to promote temperance. N. Y.

Crandall, F. M. How to keep well. 1903. 20½cm
Doubleday \$1.50 n

Aims to give definite rules and principles for avoidance of disease. Handy guide to keeping well, and treatment of such simple ailments as do not require a doctor's care. *Pub. wklly.*

Gerhard, W. P. Guide to sanitary house-inspection; or, Hints and helps regarding the choice of a healthful home in city or country. 3d ed. 1902. 19cm
Wiley \$1

Notter, J. L., & Firth, R. H. Practical domestic hygiene. New ed. 1902. 19cm (Longmans' pract. elem. sci. ser.)
Longmans 90c
Elementary human anatomy and physiology; Elementary hygiene; Domestic economy.

— & **Horrocks, W. H.** Theory and practice of hygiene (Notter and Firth) 2d ed. 1900. 24cm
Blakiston \$7 n

S Original work by J. L. Notter and R. H. Firth, published 1896.

Comprehensive, founded on E. A. Parke's *Manual of practical hygiene*.

Parkes, L. C., & Kenwood, Henry. Hygiene and public health. 2d ed. 1902. 19cm

Blakiston \$3 n

Aims to cover the whole field of sanitary science, and gives such elementary information on every topic as will enable the reader to refer with advantage to the larger textbooks. *Science*, 14: 301.

Pyle, W. L. ed. Manual of personal hygiene. 1900. 21cm
Saunders \$1.50 n

S R Full of good sense and useful information about the various powers and faculties of the body, their use and incident ailments. *Spec.* 86: 394.

Richards, Mrs E. H. (Swallow), & Talbot, Marion. ed. Home sanitation; a manual for housekeepers. Rev. ed. 1904. 17cm

Whitcomb 25c

Books of reference, p. 77-80.

Stillwell, J. S. Human hair, its care and preservation. 1900. 18cm Maple pub. co. \$1.25

Object to present in easily intelligible form facts concerning human hair, particularly those pertaining to cause, prevention and cure of conditions and diseases which result in injury and total loss. Pref.

Food

Chapin, H. D. Theory and practice of infant feeding. 1902. 23½cm
Wood \$2.25

Bibliography, p. 317-20.

Hutchison, Robert. Food and the principles of dietetics. 1903. 23cm
Wood \$3 n

S Compilation and study of modern knowledge of chemical constituents of foods and of their relation to production of energy. Information on meat juices, jellies, and extracts particularly valuable. *Sal. r.* 1901.

Pequignot, Mrs M. A. (Boland) Handbook of invalid cooking. Rev. ed. 1900. 19½cm
Century \$2

Literature, p. 313-14.

For professional and unprofessional nurses. Explanatory lessons on food properties and values, recipes, menus, with chapters on serving, feeding of children and district nursing. N. Y.

Richards, Mrs E. H. (Swallow) Food materials and their adulterations. New ed. 1898. 17½cm
Home science \$1

Bibliography, p. 175-79.

Important to housekeepers. On water, tea, coffee, dairy products, sugar, tinned goods, spices, etc. perishable foods and principles of diet. N. Y.

— & **Woodman, A. G.** Air, water and food from a sanitary standpoint. 1900. 23½cm
Wiley \$2

Elementary, with descriptions of laboratory methods for analyzing air, water and food. Tables of relative food values. Bibliography. *Engin.* n. 45: 32.

Thompson, Sir Henry. Diet in relation to age and activity. 4th ed. 1903. 18½cm

Warne \$1

Well deserves its established popularity. Instructions for attaining comfortable longevity, adapted to all intelligent people, and the more acceptable from their felicitous style. *Nation*, 74: 289.

Winthrop, A. W. Diet in illness and convalescence. 1899. 20½cm Harper \$1.50

Large variety of recipes for beverages and foods, and much practical information of miscellaneous nature. Chapter devoted to diets suitable for various specified diseases. *Lit. world*, 30: 125.

Yeo, I. B. Food in health and disease. 8th ed. 1901. 19cm Keener \$2.50 n

Nutritive values, digestion and assimilation, cooking, dietaries, animal and vegetable food, food in special diseases, etc. Comprehensive and practical. N. Y.

613.7 Hygiene of recreation and sleep

Bancroft, J. H. School gymnastics. Rev. ed. 2 v. 1900-03. 19cm

Heath v. 1, \$1.50; v. 2, \$1.75

Contents: v. 1, Free hand, rev. ed. v. 2, With light apparatus.

Systems used in Brooklyn and Waterbury, Ct. public schools.

Distinctly good. Detailed courses for boys and girls covering first 8 school years. *Educ. r.* 21: 396.

Photographic illustrations.

Benson, E. F., & Miles, E. H. Daily training. 1902. 19cm (Imp. athletic lib.)

Dutton \$1.50 n

Concerning exercise, diet, stimulants, water, heat, sleep, rest, etc. under ordinary conditions. Manly and practical. Authors agree on broad principles, differing in personal practices. N. Y.

Blaikie, William. How to get strong and how to stay so. Rev. ed. 1902. 19½cm

Harper \$1 n

Author is justifiably loyal to the exercise of walking, laying great stress on carriage, and to home gymnastics. In these respects the book is stimulating and offers a real guidance. *Nation*, 67: 370 (B)

Call, A. P. Power through repose. New ed. 1900. 17½cm Little \$1

The gospel of relaxation, as one may call it, is preached in this admirable little volume, which ought to be in the hands of every teacher and student in America of either sex. William James, *Talks to teachers* (B)

Faries, Randolph. Practical training for athletics, health and pleasure. 2d ed. 1899. 19½cm (Outing lib.) Outing \$1

American manual approved by D. A. Sargent.

Nissen, Hartvig. Rational home gymnastics, with health-points on walking and bicycling and the use of water and massage. Rev. ed. 1903. 18cm Bacon \$1

614 Public health

Abbott, A. C. Hygiene of transmissible diseases; their causation, modes of dissemination and methods of prevention. 2d ed. 1901. 24cm Saunders \$2.50 n

Places before unprofessional readers compactly and intelligibly much widely scattered matter. J. Y. Bergen in *Educ. r.* 20: 194.

Harrington, Charles. Manual of practical hygiene for students, physicians and medical officers. 2d ed. 1902. 24½cm Lea \$4.25 n

Complete, authoritative, practical and modern. Chapter on hygiene of occupation. *Science*, 37: 730.

Prepared by specialists for practical aid of communities and individuals. Points to a final triumph over the scourge of consumption through persistent, intelligent effort. N. Y.

New York charity organization society. Handbook on the prevention of tuberculosis. 1903. 24cm Charity org. soc. \$1

Sedgwick, W. T. Principles of sanitary science and the public health. 1902. 22½cm

Macmillan \$3 n

With special reference to causes and prevention of infectious diseases. Not too technical for intelligent readers. N. Y.

Sternberg, G. M. Infection and immunity. 1903. 21½cm (Sci. ser.) Putnam \$1.75 n

Pt. 1, General discussion of causes of infection, use of disinfectants, and immunity; pt. 2 treats specially of origin, death rate and method of prevention of infectious diseases. For non-medical readers.

Sykes J. F. J. Public health problems. 1892. 19cm (Contemp. sci. ser.) Scribner \$1.50

Tyndall, John. Essays on the floating-matter of the air, in relation to putrefaction and infection. 1902. 20½cm Appleton \$1.50

See also Public documents list.

614.3 Adulterations

Blyth, A. W. & M. W. Foods: their composition and analysis; with introd. on the history of adulteration. 5th ed. 1903. 23cm

Van Nostrand \$7.50

As a manual of the composition and examination of human food it has a leading position. Exact reference to every authority and a series of bibliographies make easy confirmation of quotation and further study. Legal relations are to English law, but they suggest ways of right doing. *Nation* (B)

614.8 Protection of human life from accidents, casualties, etc.

Doty, A. H. Manual of instruction in the principles of prompt aid to the injured. 4th ed. 1902. 18½cm Appleton \$1.50 n

Practical, useful, up to date. Ed. sel.

Dulles, C. W. Accidents and emergencies. 5th ed. 1899. 19½cm Blakiston \$1

First published in 1880 under title *What to do first in accidents or poisoning*.

For household use. Includes chapters on electricity, accidents and poisonous flowers, and some excellent pages on domestic emergencies. *Lit. world*, 28: 379.

Hardy, W. J. Lighthouses; their history and romance. 1895. 20cm Revell \$1.25

Y Notable lighthouses and lightships of English coast, with brief history of use of lifesaving signals in England. N. Y.

Hill, C. T. Fighting a fire. 1897. 19½cm Century \$1.50

Y Account of the organization, methods, dangers and heroism of the New York fire department. Unit. Appeared in *St Nicholas*, July 1896–Oct. 1897. Pittsburg.

615 Therapeutics. Cures

Hudson, T. J. Law of mental medicine; the correlation of the facts of psychology and histology in their relation to mental therapeutics. 2d ed. 1903. 19½cm McClurg \$1.20 n

Object primarily to assist in placing mental therapeutics on a firmly scientific basis, and incidentally to place within reach of the humblest intellect the most effective methods of healing by mental processes. *Pub. wckly*.

616 Pathology. Diseases. Treatment

Frankland, Mrs G. C. (Toynbee) Bacteria in daily life. 1903. 19½cm Longmans \$1.75 n

P R Treated in a general rather than a detailed manner. Author assumes only as much knowledge as the average educated man may safely be considered to possess. *Ath.* '03, 2: 97.

Gardiner, C. F. Care of the consumptive; with chapter on Colorado as a resort for invalids. 1900. 18cm Putnam \$1.25

Bibliography, p. 168–72.

Aims to give clear, practical rules governing use of fresh air, sunlight, food, rest and exercise. Not intended to supplant medical advice. N. Y.

Mitchell, S. W. Fat and blood; an essay on treatment of certain forms of neurasthenia and hysteria; 8th ed. with additions by J. K. Mitchell. 1902. 19½cm Lippincott \$1.50

Muir, Robert, & Ritchie, James. Manual of bacteriology; Amer. ed. rev. by N. M. Harris. 1903. 23cm Macmillan \$3.75 n

Bibliography, p. 534–52.

Clear and well developed exposition. Includes remarkable discoveries of Reed and his colleagues about yellow fever. Much larger and more complete than the excellent first edition (1897) *Nation*, 76: 331.

617 Surgery

Howe, J. W. Emergencies; etiology, pathology and treatment of accidents, diseases and cases of

poisoning which demand prompt action. 4th ed. 1884. 23cm

S

Appleton \$2.50

618 Obstetrics

Davis, E. P. Mother and child. 2d ed. 1902. 19½cm Lippincott \$1.50 n

Not intended to take the place of the family physician, but to help him in care of his patients by placing at their disposal information which physicians commonly give to their patients. *Introd.*

619 Veterinary medicine

Mayo, N. S. Care of animals. 1903. 18cm (Rural sci. ser.) Macmillan \$1.25 n

P Takes up surgery, medicine and obstetrics in relation to farm animals and admirably translates the technical into the simple and intelligible. Contains directions concerning general care and quarters of farm animals, care of pets and the judging, handling and shoeing of horses. *Nation*, 78: 130.

See also Public documents list.

620 Engineering

Burr, W. H. Ancient and modern engineering and the Isthmian canal. 1902. 23½cm Wiley \$3.50 n

Expansion of 6 Cooper union lectures. Concise, accurate accounts of Nicaragua and Panama canals. Written with good judgment and full knowledge of latest thought and research. *Engin. n. v.* 49, Jan. 15, sup. p. 8.

Carpenter, R. C. Experimental engineering and manual for testing. 5th ed. 1903. 23½cm Wiley \$6

S Standard methods for testing materials, engines and machinery. Aim not to teach how to generalize safely from experiments, but rather how to make the experiments themselves. Much of the information is found elsewhere only in scattered papers and articles. *Engin. n.* 28: 202.

Cochrane, C. H. Wonders of modern mechanism. 3d ed. 1900. 20cm Lippincott \$1.50 n

P The trolley, electric light, 30-story building, horseless carriage, kinetoscope, and other marvels. Clear, forcible English. Providence.

Haswell, C. H. Mechanics' and engineers' pocket-book of tables, rules and formulas pertaining to mechanics, mathematics and physics. 69th ed. 1903. 16½cm Harper \$4

Rf Including areas, squares, cubes, roots, logarithms, hydraulics, hydrodynamics, steam and steam engine, naval architecture, masonry, steam vessels, mills, limes, mortars, cements, etc. Full index. N. Y.

Johnson, J. B. Engineering contracts and specifications. 3d ed. 1902. 23cm Engin. n. \$3

S Contains synopsis of law of contracts and complete list of descriptive and technical clauses of various kinds

of engineering specifications, for students, engineers and contractors; specifications complete and carefully worded. *Elec. world*, 32: 634.

Johnson, J. B. Materials of construction. 2d ed. 1898. 23½cm Wiley \$6

S Main results of foreign investigators. Indicates where details of any particular line of work may be found. N. Y.

Lane, Mrs M. A. L. ed. Triumphs of science. 1903. 18cm (Youths' companion ser.)

Y Ginn 30c

Merriman, Mansfield. Strength of materials; text-book for manual training schools. 1897. 19½cm Wiley \$1

Patton, W. M. Treatise on civil engineering. 2d ed. 1903. 24cm Wiley \$7.50

S Bibliography, pref. p. 18.

Aims to present essential and fundamental principles and a selection of practice without attempting exhaustive treatment. *Engin. n.* 34: 73.

Thurston, R. H. Materials of engineering. 3 v. 1899-1903. 23½cm Wiley \$8

Contents: pt. 1, Non-metallic materials of engineering: stone, timber, fuel, lubricants, etc. 5th ed. rev. 1899. pt. 2, Iron and steel. 9th ed. rev. 1903. pt. 3, Brasses, bronzes, and other alloys, and their constituent metals. 4th ed. rev. 1900.

Trautwine, J. C. Civil engineer's pocket-book; rev. by J. C. Trautwine jr. and J. C. Trautwine 3d. 18th ed. 1902. 17½cm Wiley \$5

Rf Bibliography, p. 1008-23.

Perhaps the most consulted reference work for civil and mechanical engineer. Tables trustworthy and of great intrinsic value. *Sci. Amer.* 88: 165.

Unwin, W. C. The testing of materials of construction; a text-book for the engineering laboratory and a collection of the results of experiment. 2d ed. 1899. 22½cm Longmans \$6 n

621 Mechanical engineering

Archbutt, Leonard, & Deeley, R. M. Lubrication and lubricants; a treatise on the theory and practice of lubrication, and on the nature, properties and testing of lubricants. 1900. 22½cm Lippincott \$5.50 n

Barber, T. W. Engineer's sketch-book of mechanical movements, devices, appliances, contrivances and details employed in the design and construction of machinery. 4th ed. 1902. 22cm Spon \$4

Text consists of diagrams, with descriptive notes on opposite pages.

— Repair and maintenance of machinery; a handbook of practical notes and memoranda. 1895. 22½cm Spon \$3.50

Barr, J. H. Kinematics of machinery; a brief treatise on constrained motions of machine elements. 1899. 23½cm Wiley \$2.50

Elucidates and supplements Unwin's *Machine design*. Craver.

Barr, W. M. Pumping machinery; a practical hand-book relating to the construction and management of steam and power pumping machines. 1893. 24cm Lippincott \$5

Details plainly described and well illustrated; with caution and advice evidently derived from practical knowledge. *Engin. n.* 29: 326.

Blaine, R. G. Hydraulic machinery, with an introd. to hydraulics. 1897. 22cm (Finsbury tech. man.) Spon \$5

Theoretic and practical data under each head, with full illustrations of late applications. *Engin. n.* 38: 128.

Bodmer, G. R. Hydraulic motors and turbines. 3d ed. 1902. 19½cm (Specialists' ser.)

Van Nostrand \$5

Gives theory and practice of turbine construction, describes various systems in use and typical installations. Requires considerable knowledge of mathematics. Pittsburgh.

Brown & Sharpe manufacturing co. Providence, R. I. Practical treatise on gearing. 7th ed. 1902. 23cm Brown \$1

Compton, A. G., & De Groodt, J. H. Advanced metal-work: pt. 1, The speed-lathe. 1898. 19cm Wiley \$1.50

A practical instruction book in hand-turning in wood and to a lesser extent in brass. *Amer. mach.* 22: 290 (B)

Durand, W. F. Practical marine engineering. 1901. 23½cm Marine engin. \$5

Authoritative and encyclopedic; subject reduced to its simplest and most exact terms. R. H. Thurston in *Science*.

Frizell, J. P. Water power; an outline of the development and application of the energy of flowing water. 3d ed. 1903. 23½cm

Wiley \$5

Comprehensive and up-to-date; in no sense elementary. New practice plainly set forth and illustrated by examples from existing power plants. *Engin. n.* 45: 79.

Goodman, John. Mechanics applied to engineering. 1899. 19cm Longmans \$2 n

Much condensed, but clear and logical. *Engin. n.* 42: 312.

Hiscox, G. D. Compressed air; its production, uses and applications. 1901. 25cm

Henley \$5

Compilation of all sorts of information from books, papers and trade catalogs presented in a semi-popular, semi-technical manner. *Engin. n.* 48: 30.

- Horner, J. G.** Principles of fitting; by a foreman pattern maker. 1893. 18cm
Macmillan \$1.50
Directs attention to cardinal matters at basis of the trade. Pref.
- Hutton, F. R.** Gas-engine; a treatise on the internal-combustion engine using gas, gasoline, kerosene or other hydrocarbon as source of energy. 1903. 23½cm Wiley \$5
Bibliography, p. 467-69.
Describes American types. Craver.
- Kent, William.** Mechanical engineer's pocket-book; a reference-book of rules, tables, data and formulae. 6th ed. 1903. 17½cm Wiley \$5
Aims to cover mechanical engineering as thoroughly as Trautwine covers civil engineering. Notable feature is quotation of authority on which rules, tables, formulas, etc. are based. *Engin. n.* 33: 328.
- Lineham, W. J.** Text-book of mechanical engineering. 5th ed. 1902. 21cm Lippincott \$4.50 n
Pt. 1, Workshop practice, contains general operations: pattern making, molding, forging, machine tools, fitting and erecting, etc. pt. 2, Theory and examples: strength of materials and structures, energy and transmission of power, heat and heat engines, hydraulics and hydraulic machines. 732 drawings in text, 18 folding plates. Good index. *Ath.* '95, 2: 495.
- Low, D. A., & Bevis, A. W.** Manual of machine drawing and design. Rev. ed. 1902. 22cm Longmans 75c n
Text concise and plain, illustrated by dimensioned shop drawings. Adapted for students but also full of practical workshop hints and suggestions. *Engin. n.* 29: 463 (B)
- Roberts, E. W.** Gas-engine handbook. 4th ed. 1903. 13½cm Gas engine \$1.50
Details of principle of internal combustion engines, various types, fuels and heating values, operations and care of gas and gasoline engines, internal combustion engines, building and designing of gas engine. Requires knowledge of simple equations in algebra. *Elcc. world*, 35: 799.
- On marine motors and motor launches. 1901. 16cm (Rudder on ser.) Rudder \$1
Devoted exclusively to gasoline motor. Pittsburg.
- Robinson, S. W.** Principles of mechanism. 1903. 23½cm Wiley \$3
Aims to treat whole subject of mechanism so systematically and comprehensively that by its aid any machine may be analyzed into its elementary combinations, and the character of their motions determined. Pref.
- Robinson, William.** Gas and petroleum engines. 2d ed. 2 v. 1902. 22½cm (Finsbury tech. man.) Spon \$8.50
Aims to aid students and engineers in studying engines and principles that underlie and control their action. Traces evolution of internal combustion engine. Chief types of motors shown by many illustrations. Pref.
- Rose, Joshua.** Complete practical machinist. 19th ed. 1903. 20½cm Baird \$2.50
Lathe work, vise work, drills and drilling, taps and dies, hardening and tempering, the making and use of tools, tool grinding, marking out work, machine tools, etc. Title.
- Smith, A. W.** Materials of machines. 1902. 19cm Wiley \$1
Short discussion with particular attention to physical and mechanical properties of iron and copper and their alloys. Intended to aid in selecting materials for various parts of machines. Pittsburg.
- Taylor, A. J. Wallis-** Refrigeration, cold storage and ice-making. 1902. 23cm Van Nostrand \$4.50 n
Elaboration of *Refrigerating and ice-making machinery*. Appendix: Useful tables and memoranda; Bibliography of refrigeration.
Traces mechanical development of different refrigerating processes and describes machinery and appliances employed in modern and improved practice. Copiously illustrated. *Engin. n.* v. 48, 18 Dec. lit. sup. p. 4.
- Unwin, W. C.** Elements of machine design. Rev. ed. 2 v. 1901. 18cm (Text-bks. of sci.) Longmans v. 1, \$2.25; v. 2, \$2
Admirable and unique. Fairly complete account of methods of proportioning parts of machinery, specially of steam engine. *Science*, 16: 10 (B)
- Usher, J. T.** Modern machinist. 5th ed. 1904. 20cm Henley \$2.50
Compendium of shop methods, showing a variety of special tools and appliances. Ed. sel.
- Van Dervoort, W. H.** Modern machine shop tools, their construction, operation and manipulation, including both hand and machine tools. 1903. 24cm Henley \$4
Describes minutely and illustrates 673 tools and methods. N. Y.
- Wilson, H. S.** Practical tool-maker and designer: a treatise upon the designing of tools and fixtures for machine tools and metal working machinery. 1898. 23½cm Baird \$2.50
- Wood, De Volson.** Turbines, theoretical and practical. 2d ed. 1896. 24cm Wiley \$2.50
Seeks to simplify solution of turbine problems by deducing from wheels of commercial proportions certain numeric results. *Engin. n.* 33: 232.

621.1 Steam engineering

- Forney, M. N.** Catechism of the locomotive. 2d ed. 1890. 21½cm Van Nostrand \$3.50

Based on Georg Kosak's *Katechismus der locomotive*. Clear, simple description of construction and working of the locomotive engine. On many railroads the standard for examination of firemen for promotion. *Van Nostrand*, 19: 25.

- Kent, William.** Steam-boiler economy. 1901. 23½cm Wiley \$4
Eminently practical and useful. *Sci. Amer.* 85: 44 (B)
- Neilson, R. M.** Steam turbine. 2d ed. 1903. 23½cm Longmans \$3.60
Includes short history of development, sketches theory and describes in detail the De Laval, Ra'ean and Parsons types. Chapter on propulsion of ships by turbines. Pittsburgh.
- Peabody, C. H., & Miller, E. F.** Steam-boilers. 1897. 24cm Wiley \$4
S Authors professors in Mass. institute of technology. Illustrations excellent. *Engin. n.* 37: 272.
- Reagan, H. C.** Locomotives: simple, compound, and electric. 4th ed. 1902. 19cm Wiley \$2.50
Ed. 1-3 issued under title *Locomotive mechanism and engineering*.
Deals largely with handling and emergency repairing. *Engin. n.* 36: 304 (B)
- Reeve, S. A.** Thermodynamics of heat-engines. 1903. 19½cm Macmillan \$2.60 n
Contents: Theory; Application of theory to practice.
Appended: The steam-table, comp. from various sources by S. A. Reeve, 1903, 42 p. of tables, 35c n.
- Ripper, William.** Steam-engine theory and practice; with 438 illustrations. 1899. 22½cm Longmans \$2.50
Sequel to author's elementary book *Steam*. Pref.
Most useful portions to American engineers are those which give the applications of thermodynamic analysis to actual engine conditions. *Amer. mach.* 23: 97 (B)
- Spangler, H. W. & others.** Elements of steam engineering. 1903. 23½cm Wiley \$3
Intended to bring before beginner examples of various forms of apparatus used in modern steam power plants, to explain simply and briefly construction, use and reasons for using these various parts or machines, and to give him a working vocabulary. Pref.
- Thurston, R. H.** History of the growth of the steam-engine. 4th ed. 1897. 19½cm (Internat. sci. ser.) Appleton \$2.50
Manual of steam-boilers. 7th ed. 1901. 23½cm Wiley \$5
S Exhibits fully progress of boiler in construction and adaptability for various purposes. Chapter on operation and care of boilers clear, concise and worthy of close attention. *Van Nostrand*, 16: 35.
- Manual of the steam engine. 5th ed. 2 v. 1900-02. 23½cm Wiley pt. 1, \$6; pt. 2, \$7.50
Contents: Structure and theory; Design, construction and operation.
For educated, practical engineers and advanced technical students. Combines long established thermodynamic theory of the heat engine with wastes in practical use, or losses of heat and energy, and thus establishes complete theory of the machine, its operation and efficiency. *Engin. n.* 26: 212.
- 621.3 Electric engineering**
- Atkinson, A. A.** Electrical and magnetic calculations. 2d ed. 1903. 19cm Van Nostrand \$1.50 n
Covers the definite theory of electricity and magnetism and enforces each phase with admirably clear and practical computation. Numerous problems illustrating text fully worked out; judiciously selected problems for solution. Exhaustive index. *Elec. world* 40: 142.
- Bell, Louis.** Electric power transmission. 3d ed. 1901. 24cm McGraw \$3
R Aims to set forth as simply as possible present practice. For business man rather than student; treatment eminently practical. *Engin. n.* 37: 192.
Value appreciably increased by addition of short chapter on recent achievements in high-voltage transmission. *Engin. n.* 43: 63.
- Power distribution for electric railroads. 3d ed. 1900. 24cm Street railway pub. co. \$2.50
Important problems connected with design of system of conductors for electric railways fully considered in light of existing conditions and satisfactory solutions reached. *Elec. world*, 29: 296.
- Crocker, F. B.** Electric lighting. 2 v. 1896-1901. 24cm Van Nostrand \$3 ea
Contents: v. 1, Generating plant. v. 2, Distributing system and lamps.
Practical exposition of the art for engineers, students and others interested in installation or operation of electric plants. Title.
- & Wheeler, S. S. Practical management of dynamos and motors. 5th ed. 1903. 19½cm Van Nostrand \$1
Originally published in *Electrical engineer*, Sep. 1891-May 1892.
Practical treatise for engineers and an excellent hand-book. Gives careful directions for examining and testing generators, also for localization and remedy of troubles in all kinds of dynamos and motors. E. ass'n. of phy. teachers (B)
- Cushing, H. C. jr.** Standard wiring for electric light and power, as adopted by the fire underwriters of the United States; containing national electrical code explained and illustrated. 1903. 17½cm Author, New York \$1
- Foster, H. A.** Electrical engineer's pocket-book. 2d ed. 1902. 17½cm Van Nostrand \$5
Author practising engineer of reputation. Compend of useful facts for daily use of electrician and electric engineer. *Engin. n.* 46: 469.
- Hanchett, G. T.** Modern electric railway motors; a discussion of current practice in electric railway motor construction, maintenance and repair. 1900. 23cm Street railway pub. co. \$2
Contains table of principal dimensions of all types of street railway motors now in general use, American and foreign. *Elec. world*, 35: 878.

Herrick, A. B. Practical electric railway handbook. 1901. 17½cm Street railway pub. co. \$3

Construction, testing and operating of electric railway systems, according to best American practice. Avoids mathematics as much as possible, and illustrates by cuts and diagrams appliances and methods proposed. *Engin. n.* 46: 437.

Houston, E. J., & Kennelly, A. E. Alternating electric currents. 3d ed. 1902. 17½cm (Elem. electro-tech. ser.) McGraw \$1

Fundamental principles necessary to correct understanding of subject developed in simplest language and without any mathematics except arithmetic. Practical applications described in terms as free as possible from technicalities. *Elec. world*, 26: 653.

Electric arc lighting. 2d ed. 1902. 17½cm (Elem. electro-tech. ser.) McGraw \$1

Mechanism of different types of lamps shown, giving good idea of present practice. *Elec. world*, 28: 341.

Electric heating. 1895. 17½cm (Elem. electro-tech. ser.) McGraw \$1

Covers subject very thoroughly, though not a general treatise. Chapters on elementary principles, heating of bare conductors, covered conductors, fuse wires, electric heaters, electric cooking, electric welding, electric furnaces, miscellaneous applications. *Elec. world*, 27: 321.

Electric incandescent lighting. 2d ed. 1902. 17½cm (Elem. electro-tech. ser.) McGraw \$1

Brief history of early art of electric lighting, description of arc light generators, lamp mechanism, and various processes in manufacture of carbon electrodes. St. Louis.

The electric motor and the transmission of power. 1896. 17½cm (Elem. electro-tech. ser.) McGraw \$1

Electric street railways. 1896. 17½cm (Elem. electro-tech. ser.) McGraw \$1

Sketch of history, brief statement of elementary principles, descriptions of parts of an electric railway system, and chapters on storage battery, traction and electric locomotives. Non-technical. *Elec. world*, 28: 764.

Kapp, Gisbert. Transformers for single and multiphase currents; a treatise on their theory, construction, and use; with 133 illustrations. 1896. 19½cm (Specialists' ser.) Macmillan \$1.75

Perrine, F. A. C. Conductors for electrical distribution; their materials and manufacture, the calculation of circuits, pole-line construction, underground working, and other uses. 1903. 24cm Van Nostrand \$3.50 n

Gives in authoritative and systematic form the benefit of extensive experience. For student and practising engineer. *Engin. n.* v. 49, 16 Ap. lit. sup. p. 42-43.

Poole, C. P. Electric wiring. 1900. 17cm Power \$1

Assumes merely familiarity with Ohm's law. For practical wiremen and electric engineers. *Elec. world*, 35: 447.

Rosenberg, E. Electrical engineering; tr. by W. W. H. Gee and Carl Kinzbrunner. 1903. 23½cm Wiley \$1.50 n

Practical [non-mathematical] Explains construction of machines and how construction affects running; how speed, voltage, load, torque, etc. vary; what troubles arise from reversed connections; how a machine will behave under special conditions, etc. *Engin. n.* lit. sup. 50: 15.

Sheldon, Samuel, & Mason, Hobart. Dynamo electric machinery; its construction, design and operation. 2 v. 1901-02. 20cm Van Nostrand \$2.50 ea n

v. 1, Direct current machines, 2d ed. v. 2, has title *Alternating-current machines*.

Shepardson, G. D. Electrical catechism; an introductory treatise on electricity and its uses. 1901. 26cm Amer. elec. co. \$2

Non-technical.

Sloane, T. O'C. How to become a successful electrician. 12th ed. 1903. 18½cm Henley \$1

Containing studies to be followed, methods of work, field of operation, professional ethics, etc. Title.

Thompson, S. P. Design of dynamos. 1903. 23½cm Spon \$3.50

622 Mining engineering

Foster, C. Le N. Text-book of ore and stone mining; 4th ed. by L. H. Cooke. 1901. 22cm Lippincott \$10 n

Greene, Homer. Coal and the coal mines. 1889. 17cm (Riverside lib. for young people) R Y Houghton 75c

Shinn, C. H. Story of the mine. 1896. 19½cm (Story of West ser.) Appleton \$1.50

Mining conditions as illustrated by history of Comstock lode, with side chapters on mine litigation, speculation, mechanical problems, etc. Picturesque and authoritative. N. Y.

623 Military and naval engineering

Fyfe, H. C. Submarine warfare, past, present and future; with introd. by Admiral Tremantle, and chapter on the probable future of submarine boat construction by Sir E. J. Reed. 2d ed. 1903. 22½cm Dutton \$3 n

P Short bibliography, p. 331-32.

Sociologic and metaphysical sides discussed with animation. Reviews American invention and enterprise. Much technical data and admirably illustrated. *Elec. world*, 40: 426.

624 Bridges and roofs

Baker, I. O. Treatise on masonry construction. 9th ed. 1899. 23½cm Wiley \$5

Most valuable and complete treatise on masonry in any language. *Engin. n.* 42: 211.

Howe, M. A. Design of simple roof-trusses in wood and steel; with an introd. to the elements of graphic statics. 1902. 23½cm Wiley \$2

Useful to those without special training in structural engineering. Includes tables giving the properties of standard steel shapes. Craver.

Johnson, J. B. & others. Theory and practice of modern framed structures; for schools, and for engineers in professional practice. 8th ed. 1904. 28 x 22cm Wiley \$10

Maginnis, O. B. Roof framing made easy. 2d ed. 1903. 21cm Industrial pub. co. \$1.50

Reprint of articles in *Carpentry and building* and the *Carpenter*.

Gives simple, practical method for laying out and framing timber roofs of various forms. Concise; illustrations profuse and clear. *Engin. n.* 41: 111 (ed. 1)

625 Railroad and road engineering

Baker, I. O. Treatise on roads and pavements. 1903. 23½cm Wiley \$5

Discussion of principles from engineer's point of view. Considers economics and location of country roads, construction and maintenance of earth roads, hard surface roads and pavements. Pittsburg.

Homans, J. E. Self-propelled vehicles. 1902. 23cm Audel \$5

Practical treatise on theory, construction, operation, care and management of all forms of automobiles. Title.

Shaler, N. S. American highways. 1896. 19cm Century \$1.50

P List of important works on highway construction, p. 292-93.

Road materials, methods of construction and administration, machinery, cost, effects of climate and topography, importance of specific training and education in highway engineering, etc. N. Y.

Spalding, F. P. Text-book on roads and pavements. 2d ed. 1902. 19cm Wiley \$2

Tillson, G. W. Street pavements and paving materials. 1900. 23½cm Wiley \$4

Comprehensive, authoritative review of best current practice. Illustrations. *Engin. n.* 45: 32.

Tratman, E. E. R. Railway track and track work; 2d ed. with sup. chapters on "Signals and interlocking" and "Street railway track." 1901. 22½cm Engin. n. \$3

General principles with numerous details of material, appliances and work, and methods of practice, with drawings. Various subjects treated critically and descriptively. Pref.

Warman, Cy. Story of the railroad. 1898. 19½cm (Story of west ser.) Appleton \$1.50

Picture of the life of the man who built the trans-continental lines across the plains and mountains. Also, a history, in brief, of the five principal companies whose roads have crossed the continent. *Lit. world* (B)

Webb, W. L. Railroad construction: theory and practice; a text-book for the use of students in colleges and technical schools. 2d ed. 1903. 17½cm Wiley \$5

Railway work from surveys to track construction. *Engin. n.* 45: 7.

Wellington, A. M. Economic theory of the location of railways; an analysis of the conditions controlling the laying out of railways to effect the most judicious expenditure of capital. Rev. ed. 1887. 21½cm Wiley \$5

626 Canal engineering

Mead, Elwood. Irrigation institutions. 1903. 19½cm (Cit. lib. of econ. pol. & sociol.)

Macmillan \$1.25 n

Discussion of the economic and legal questions created by growth of irrigated agriculture in West. Title.

627 River, harbor and general hydraulic engineering

Harcourt, L. F. V. Harbours and docks. 2 v. 1885. 23cm (Clarendon press ser.)

Clarendon \$6.50

Their physical features, history, construction, equipment and maintenance, with statistics as to their commercial development. Title.

— Rivers and canals. 2d ed. 2 v. 1896. 23cm Clarendon \$8

Contents: v. 1, Rivers. v. 2, Canals.

The flow, control and improvement of rivers and the design, construction and development of canals both for navigation and irrigation, with statistics of the traffic on inland waterways. Title.

Thomas, B. F., & Watt, D. A. Improvement of rivers. 1903. 30½cm Wiley \$6 n

Principal works consulted, pref. p. 5-6.

Treatise on the methods employed for improving streams for open navigation, and for navigation by means of locks and dams. Title.

Treats also of improvement for flood protection. Authors have had charge of some of most important recent slack water navigation improvements. *Engin. n.* v. 49, 14 May, lit. sup. p. 59.

Wegmann, Edward. Design and construction of dams; including masonry, earth, rock-fill, and timber structures, also the principal types of movable dams. 4th ed. 1899. 30½ x 24½cm Wiley \$5

628 Sanitary engineering

Baker, M. N. Municipal engineering and sanitation. 1902. 19½cm (Cit. lib. of econ. pol. & sociol.) Macmillan \$1.25 n

P Practical, authoritative discussion of problems of transportation, communication, water, lighting, slaughter houses, sewage, street cleaning, smoke, noise, etc. public policy and finance. N. Y.

Author's wide practical experience furnishes many tangible examples. Brooks.

Billings, J. S. Ventilation and heating. 1893. 24cm McGraw \$6

S For architects, engineers and non-professional inquirers. Comprehensive discussion of principles and practical applied science. N. Y.

Folwell, A. P. Sewerage; the designing, construction and maintenance of sewerage systems. 5th ed. 1904. 23½cm Wiley \$3

Gerhard, W. P. House-drainage and sanitary plumbing. 10th ed. 1902. 15cm (Van Nostrand sci. ser.) Van Nostrand 50c

Reprinted from *Van Nostrand's magazine*.

— Sanitary engineering. 1898. 19½cm Author, New York \$1.25

Reprint of Franklin institute address, 1895. Indicates broad, general principles. Semi-popular. *Engin. n.* 39: 248.

Goodell, John. Water-works for small cities and towns. 1899. 23cm (Engin. rec. ser.) McGraw \$2

Refers to successful methods; not technical, of service to trustees as well as executive officers. Ed. sel.

Goodrich, W. F. Economic disposal of towns' refuse. 1901. 22cm (Engin. times lib.) Wiley \$3.50

More than a description of practice; discusses principles and criticizes. Strong bias in favor of cremation. *Engin. n.* 46: 198.

Hill, J. W. Purification of public water supplies. 1898. 23½cm Van Nostrand \$3

Sums up case against impure water and the remedy. Discusses quality of water supplies and relation to typhoid fever; sedimentation, sterilization and filtration; description of American and foreign filter plants. *Engin. n.* 39: 296.

Mason, W. P. Water-supply. 3d ed. 1902. 23½cm Wiley \$4

Omits chapters on chemical and bacteriologic examination, now published separately. *Engin. n.* 48: 52.

Merriman, Mansfield. Elements of sanitary engineering. 2d ed. 1899. 23½cm Wiley \$2

Adapted for college use. List of topics for further investigation at end of each chapter. *Engin. n.* 40: 424.

Rafter, G. W., & Baker, M. N. Sewage disposal in the United States. 1894. 26½cm Van Nostrand \$6

Rideal, Samuel. Sewage and the bacterial purification of sewage. 2d ed. 1901. 23½cm Wiley \$3.50

Best on subject. Burgess.

First comprehensive and satisfactory work on sewage purification since such bacterial methods as septic tank and rapid filter beds. Judicial rather than partizan. *Engin. n.* 45: 33.

Waring, G. E. How to drain a house. 2d ed. 1895. 17½cm Van Nostrand \$1.25

Practical expert information for individual householder. N. Y.

— Modern methods of sewage disposal for towns, public institutions and isolated houses. 1894. 19½cm Van Nostrand \$2

Expert, practical discussion of general problems and methods, sewage irrigation, filtration, chemical treatment, needs of hotels, country houses, etc. N. Y.

— Sanitary condition of city and country dwelling houses. 2d ed. 1898. 15½cm (Van Nostrand sci. ser.) Van Nostrand 50c

Papers on country and city houses, and correspondence regarding them in *American architect*. States some essentials of house sanitation. *Engin. n.* 41: 372.

630 Agriculture

Brooks, W. P. Agriculture. 3 v. 1901. 22cm King \$2.75

Contents: v. 1, Soils, formation, physical and chemical characteristics and methods of improvement, including tillage, drainage and irrigation. v. 2, Manures, fertilizers and farm crops, including green manuring and crop rotation. v. 3, Animal husbandry, including breeds of live stock, general principles of breeding, feeding animals; discussion of ensilage, dairy management on farm and poultry farming.

Burkett, C. W: & others. Agriculture for beginners. 1903. 19cm Ginn 75c

Contents: Soil; The soil and the plant; The plant; How to raise a fruit tree; Diseases of plants; Orchard, garden and field insects; Farm crops; Domestic animals; Farm dairying.

Conn, H. W: Agricultural bacteriology. 1901. 21½cm Blakiston \$2.50 n

Study of the relation of bacteria to agriculture, with special reference to the bacteria in the soil, in water, in the dairy, in miscellaneous farm products and in plants and domestic animals. Title.

Gregg, Mrs Mary (Kirby), & Kirby, Elizabeth. Aunt Martha's corner cupboard; or, Stories about tea, coffee, sugar, rice, etc. 1898. 19½cm Flanagan 40c

Roberts, I. P. The farmstead. 1900. 18cm (Rural sci. ser.) Macmillan \$1.25

On selection and purchase of farms and location of buildings. Chapters on household administration, water supply, sewage, etc. Clear cut and practical. *Nat. nurseryman*.

Woll, F. W. Handbook for farmers and dairy-men. 2d ed. 1900. 17½cm Wiley \$1.50
 Compend with tables, formulas, receipts, statistics, etc. Compiled largely from publications of experiment stations, scientific divisions of United States department of agriculture, farm papers and standard works. Includes full summary of main results of 12th census as pertaining to agricultural interests. Pref.

See also Public documents list.

631 Soil. Fertilizers. Drainage

Hall, A. D. The soil; an introd. to the scientific study of the growth of crops. 1903. 20½cm Murray 36

Bibliography, p. 277-79.

Deals with origin of soils, their physical and chemical properties and composition, methods of analysis, the living organisms within the soil, causes of fertility and sterility, soil types and the natural flora belonging to each. Original and suggestive. *Nature*, 68: 52.

King, F. H. Irrigation and drainage. 1899. 18cm (Rural sci. ser.) Macmillan \$1.50

Deals with relations of water to soils and plants, presenting practical problems with underlying principles. Avoids engineering problems. N. Y.

Newell, F. H. Irrigation in the United States. 1902. 19½cm (Lib. of econ. & pol.) Crowell \$2 n

Geographic conditions, methods of storing and distributing water, and results. But slightly technical, of practical value to farmer, abundantly illustrated. Author hydraulic engineer, U. S. geological survey. N. Y.

Roberts, I. P. Fertility of the land. 1897. 18cm (Rural sci. ser.) Macmillan \$1.25

Summary sketch of the relationship of farm practice to the maintaining and increasing of the productivity of the soil. Title.

Storer, F. H. Agriculture in some of its relations with chemistry. 7th ed. 3 v. 1897. 21½cm Scribner \$5 n

Not only is agriculture dealt with in its relations to chemistry but very largely in its relations to physics, physiology, botany and zoology. Of very great value to students and farmers alike. E. B. Voorhees in *Science*, n. s. 6: 374 (B)

Voorhees, E. B. Fertilizers; the source, character and composition of natural, home-made and manufactured fertilizers. 1898. 18cm (Rural sci. ser.) Macmillan \$1

632 Pests. Hindrances

Lodeman, E. G. Spraying of plants. 1896. 18cm (Rural sci. ser.) Macmillan \$1

Successive account of history, principles and practice of the application of liquids and powders to plants for the purpose of destroying insects and fungi. Title

Massee, George. Text-book of plant diseases caused by cryptogamic parasites. 2d ed. 1903. 19½cm Macmillan \$1.60 n

Smith, J. B. Economic entomology for the farmer and fruit-grower. 1896. 21cm P Lippincott \$2.50

633 Grains. Grasses. Fibers

Beal, W. J. Grasses of North America. 2 v. 1887-96. 23cm Holt v. 1, \$2.50 n; v. 2, \$5 n

Bibliography, 2: 671-75.

For farmers and general readers who have not studied botany. N. Y.

Myrick, Herbert. American sugar industry. 1899. 25cm Judd \$1.50

P In four parts; pt. 2, The cane sugar industry, by W. C. Stubbs.

Practical manual on production of sugar beets and sugar cane and on manufacture of sugar. Title.

Shaw, Thomas. Forage crops other than grasses; how to cultivate, harvest and use them. 1900. 19cm Judd \$1

634 Fruits. Orchards. Vineyards

Bailey, L. H. Principles of fruit-growing. 1897. 18cm (Rural sci. ser.) Macmillan \$1.25

Modern arts of growing, storing and marketing fruits on a large scale. Buffalo.

Card, F. W. Bush-fruits. 1898. 18cm (Rural sci. ser.) Macmillan \$1.50

Horticultural monograph of raspberries, blackberries, dewberries, currants, gooseberries and other shrublike fruits. Title.

Botany, treatment, diseases and specially persecuting insects of each fruit very fully and carefully treated. N. Y.

Fuller, A. S. The nut culturist. 1896. 19cm Judd \$1.50

History, varieties, methods of propagation and culture, diseases, properties and uses of the almond, beechnut, castanopsis, chestnut, filbert, hickorynut, walnut and other nut trees adapted to climate of United States. N. Y.

Gifford, J. C. Practical forestry. 1902. 19½cm Appleton \$1.20 n

For beginners in forestry, agricultural students, woodland owners, and others desiring a general knowledge of the nature of the art. Title.

Green, S. B. Principles of American forestry. 1903. 19cm Wiley \$1.50

List of best books on forestry, p. 318-19.

Elementary. Treats tree growth, the forest, forest influences, planting, propagation, forest regeneration, nursery practice, protection, rate of increase, mensuration, uses of wood, durability, forest economies, etc. N. Y.

Thomas, J. J. American fruit culturist; 21st ed. by W. H. S. Wood. 1903. 22½cm Wood \$2.50 n

Over 800 illustrations.

Contains most available accounts of varieties of fruits. Bailey.

- Waugh, F. A.** Fruit harvesting, storing, marketing. 1901. 18½cm Judd \$1
 Practical guide. Working illustrations.
See also Public documents list.

635 Kitchen garden

- Bailey, L. H.** Plant-breeding. 1895. 18cm
 (Garden-craft ser.) Macmillan \$1
 Lucid account of various methods adopted by gardeners to secure new and improved varieties. Deals with subjects of variation as influenced by physical environment, the struggle for life, the division of labor, etc. *Ath.* '96, 2: 261.

- Principles of vegetable-gardening. 1901. 18cm (Rural sci. ser.) Macmillan \$1.25
 American books on vegetable gardening, p. 244-63; Recent experiment station publications relating to vegetable gardening, p. 263-70.
 As indispensable for householder as for practical market gardener. *Chic. trib.*

- Henderson, Peter.** Gardening for profit. 1903. 19cm Judd \$1.50
 Marked an era in American horticulture. The secret of its success and of the author's was invention of new methods adapted to operations on a large scale. *Cycl. of Amer. hort.* 2: 729 (B)

636 Domestic animals

- Coburn, F. D.** Swine husbandry. 3d ed. 1903. 19cm Judd \$1.50
 Practical manual, with suggestions as to the prevention and treatment of diseases. Title.

- Henry, W: A.** Feeds and feeding. 1898. 23½cm Caspar \$2 n
 Excellent authoritative handbook for stockmen and students.

- Jordan, W. H.** Feeding of animals. 2d ed. 1903. 17½cm (Rural sci. ser.) Macmillan \$1.25 n

Account of chemical constituents of plants and animals, processes of nutrition and digestion, functions of food in the body, description of different cattle foods, and the most recent results of scientific investigation and farm practice. Intended for reader with some scientific knowledge. Pittsburg.

- Miller, Mrs Harriet (Mann)** Our home pets; how to keep them well and happy, by Olive Thorne Miller. 1894. 17½cm Harper \$1.25
 On characteristics and proper care of various kinds of birds, dogs, cats, and such unusual pets as the raccoon, alligator, etc. N. Y.

- Shaler, N. S.** Domesticated animals: their relation to man and to his advancement in civilization. 1895. 23½cm Scribner \$2.50

We have seldom seen animals so fairly treated and have never known them to be more reliably characterized. *Nation*, 62: 20 (B)

- Shaw, Thomas.** Animal breeding. 1903. 19cm Judd \$1.50
Stoddard, H. H. New egg farm. 1900. 20cm Judd \$1

Management of poultry on a large scale for commercial purposes. Title.

- Watson, G: C.** Farm poultry. 1901. 18cm (Rural sci. ser.) Macmillan \$1.25 n
 Bibliography, p. 331-34.
 Gives working details on every part of subject, written in a really scientific spirit; language clear and simple. Principles that underlie good practice brought to front. *Nature*, 64: 575 (B)

637 Dairy

- Conn, H. W:** Bacteria in milk and its products. 1903. 20½cm Blakiston \$1.25 n

Partial list of recent literature, p. 291-301.

Thorough, concise discussion of action of bacteria on milk, their use in manufacture of milk products and connection between milk bacteria and distribution of certain diseases. Not too technical for general readers, though intended specially for students in dairying, health officers, sanitarians, etc. Pittsburg.

- Wing, H: H.** Milk and its products. 1897. 18cm (Rural sci. ser.) Macmillan \$1
 Gives latest results from many agricultural experiment stations; especially useful to students; contains bibliography (p. 259-67) and dairy laws. N. Y.

638 Bees

- Root, A. I.** A B C of bee culture; rev. by E. R. Root. 1903. 25cm Judd \$1.25
 Biographies of noted bee-keepers, p. 407-39.

639 Pisciculture

- Norris, Thaddeus.** American fish-culture. 1868. 19½cm Coates \$1.75
 Embracing all details of artificial breeding and rearing of trout, culture of salmon, shad and other fishes. Title.

640 Domestic economy

- Campbell, Mrs Helen (Stuart)** Household economics. 1897. 20½cm Putnam \$1.50
 References at end of each chapter.
 States the problems of family life, driving home the conviction of their importance. The organic relation of household activities with all others is made plain. C. R. Henderson in *Dial*, 22: 286 (B)

- Clark, T. M.** Care of a house. 1903. 19½cm Macmillan \$1.50 n
 Valuable handbook on the repair and maintenance of houses, for non-technical readers. Craver.

- Herrick, Mrs Christine (Terhune)** First aid to the young housekeeper. 1900. 18cm Scribner \$1
 About marketing, kitchen fire, cooking processes, tidy and sanitary housekeeping, management of servants. N. Y.

Holt, Emily. *Encyclopædia of household economy.* 1903. 21cm McClure \$1.60 n

Treats kitchen, closets, repairs and restorations, house cleaning, laundry, plumbing, sanitation, 'keeping things,' lighting, heating, water, sick room, lawns and gardens, bleaches, disinfectants, healing simples, family sewing, etc. N. Y.

Parloa, Maria. *Home economics.* 1898. 19cm Century \$1.50

Authoritative manual of household management, inclusive, accurate, useful. Well indexed and practical for the housekeeper on a small or large scale. N. Y.

Richards, Mrs E. H. (Swallow) *Cost of living as modified by sanitary science.* 1899. 19cm Wiley \$1

Attempts to indicate best methods, and the proper division of incomes of from \$1,500 to \$3,000 between food, rent, running expenses, clothing, service, the emotional and intellectual life, etc. N. Y.

641 Cookery. Gastronomy

Abel, Mrs M. W. (Hinman) *Practical sanitary and economic cooking.* 1900? 19cm

Amer. pub. health ass'n 40c

Appended: Food of children and some remarks on household management, 1900.

States clearly food values and proper methods of preparation; gives recipes, bills of fare and prices. N. Y.

Ewing, Mrs Emma (Pike) *Art of cookery.* 1899. 19½cm Merrill \$1.75

Author experienced teacher of cooking. Manual of principles: marketing; food materials; methods of cooking; mixing; seasoning; serving and garnishing. Well indexed. Buffalo.

Farmer, F. M. *Boston cooking-school cook book; rev. with an apx. of 300 recipes and an addenda of 60 recipes.* 1904. 20cm Little \$2

Food and cookery for the sick and convalescent. 1904. 20cm Little \$1.50 n

Kirkland, E. S. *Dora's housekeeping.* 6th ed. 1899. 17cm

Y McClurg 75c

Six little cooks; or, Aunt Jane's cooking class. 10th ed. 1900. 17cm McClurg 75c

Y Recipes tried by girls from 9 to 12 under careful supervision. In story form, recording a summer's experience. N. Y.

Larned, Mrs Linda (Hull) *The hostess of to-day.* 1899. 21cm Scribner \$1.50

Suggestions as to table service, formal and informal dinners, luncheons, teas, etc. Many compact recipes with approximate prices. N. Y.

Lincoln, Mrs M. J. (Bailey) *Boston school kitchen text-book.* 1887. 19cm Little \$1

Study of food and explanation of general principles in connection with practical lessons in plain cooking. Adapted to use of classes in public and industrial schools Prof.

Lincoln, Mrs M. J. (Bailey) *Mrs Lincoln's Boston cook book.* 1900. 19½cm Little \$2
Trustworthy guide in practical cookery. Arrangement of topics systematic and directions for work concise and exact. A useful outline of the chemistry and physiology of food. L. & I.

Parloa, Maria. *Miss Parloa's young housekeeper; designed especially to aid beginners.* 1894. 18½cm Estes \$1 n

Richards, Mrs E. H. (Swallow), & Elliott, S. M. *Chemistry of cooking and cleaning.* 2d ed. 1897. 17½cm Home science 50c

S Books of reference, p. 153-54.

Excellent practical manual. Assumes elementary knowledge of chemistry. N. Y.

Rorer, Mrs Sarah (Tyson) *Mrs Rorer's new cook book; a manual of housekeeping.* 1902. 21cm Arnold \$2 n

Terhune, Mrs M. V. (Hawes) *Breakfast, luncheon and tea; by Marion Harland.* 1900. 19½cm (Common sense in the household ser.) Scribner \$1.75

Marion Harland's complete cook book. 1903. 22cm Merrill \$2

A practical and exhaustive manual of cookery and housekeeping, containing thousands of carefully proved recipes. Title.

Thompson, Sir Henry. *Food and feeding.* 10th ed. 1899. 20½cm Warne \$1.75

Whitney, Mrs A. D. (Train) *Just how; a key to the cook-books.* 16cm Houghton \$1

A "little grammar of cuisine", giving details that most cookbooks assume as known: list of utensils, timetables, order and methods of mixing, measures, etc. Contains tested recipes in sufficient variety for most tables, with explicit directions for preparation. N. Y.

Williams, M. E., & Fisher, K. R. *Elements of the theory and practice of cookery; a text-book for schools.* 1901. 18½cm

Macmillan \$1 n

Williams, W. M. *Chemistry of cookery.* 1902. 19½cm Appleton \$1.50

Explains in simple terms chemistry of boiling, roasting, grilling, frying and stewing. Discusses nourishing qualities of foods and effect of cooking. Clear and authoritative. N. Y.

643 Food

Richards, Mrs E. H. (Swallow) *Cost of food; a study in dietaries.* 1901. 19cm Wiley \$1

Bibliography, p. 155-58.

Study of food requirements and economical means of meeting them for infants, school children, brain workers, the aged, etc. Working tables and menus. N. Y.

645 Furniture

French, L. H. Homes and their decoration. 1903. 22cm Dodd \$3 n

Goes into great detail in descriptions and suggestions, those concerning color being particularly good. Many illustrations. Pittsburg.

Wheeler, Mrs Candace (Thurber) Principles of home decoration. 1903. 22½cm Doubleday \$1.80 n

Deals with theory and specific applications, art being understood to imply both appropriateness and beauty. Color, fitness, kitchen, bedrooms, walls, floors, draperies, furniture, among subjects discussed. N. Y.

646 Clothing

Broughton, Mrs J. Practical dressmaking. 1902. 18½cm Macmillan 75c

Besides giving general directions for cutting and fitting, tells how to patch and do plain sewing. Olcott.

James, T. M. Longmans' complete course of needlework, knitting and cutting-out. 1901. 19cm Longmans \$2

647 Servants: training, duties, wages

Salmon, L. M. Domestic service; 2d ed. with an additional chapter on domestic service in Europe. 1901. 22½cm Macmillan \$2

S Bibliography, p. 317-22.

From economic standpoint, based on statistics gathered from employers, employees and census returns. Sketches history and suggests possible remedies. N. Y.

Springsteed, A. F. The expert waitress. 1903. 17½cm Harper \$1

Painly written directions concerning all phases of the work. Instructions for carving, serving, care of rooms, dishes and lamps, picnic and traveling lunches, invalid's room, etc. and chapters on truthfulness, adaptability and a servant's contract. Buffalo.

648 Laundry

Balderston, L. R., & Limerick, M. C. Laundry manual. 1900. 19½cm Avil 50c

649 Nursery. Children. Sickroom

Griffith, J: P. C. Care of the baby. 3d ed. 1903. 21cm Saunders \$1.50 n

Manual for mothers and nurses, containing practical directions for the management of infancy and childhood in health and in disease. Title.

Harrison, Eveleen. Home nursing. 1900. 18cm Macmillan \$1

Simplest rules and remedies to be used in care of sick, some general directions regarding nourishment, and simple recipes for invalid cooking. Pref.

Oppenheim, Nathan. Care of the child in health. 1900. 18½cm Macmillan \$1.25

A physician's practical counsels to parents. Beginning with preparations for child's advent, discusses feeding, bathing, sleep, habits, education, defective children, common diseases, relations of parents to children. N. Y.

Uffelmann, J. A. Manual of the domestic hygiene of the child; tr. by H. R. Milinowski; ed. by M. P. Jacobi. 1891. 23½cm Putnam \$1.75
Scientific and comprehensive. For trained readers. L. & I.

Wheeler, Marianna. The baby; his care and training. 1901. 17½cm Harper \$1 n
Plain and sensible in the last degree. *Nation*, 74: 289 (B)

650 Communication. Commerce**652 Writing**

Jackson, John. Theory and practice of handwriting. Rev. ed. 1894. 21cm Harison \$1

Claims of vertical writing, forms of letters, and directions for teaching. Brief history of former use of upright handwriting, its decay and revival. L. & I.

653 Abbreviations. Shorthand

Kimball, Duran. Business shorthand; presenting a method of swift writing for the use of amanuenses and reporters in accordance with the principles of Lindsley's takigrafiy. 3d ed. 1902. 20cm Kimball \$1

Kitt, C: W. Shorthand dictionary; containing characters written in accordance with lessons in Pitman phonography. 1901. 15½ x 7½cm

Rf Powers \$1

Munson, J. E. Art of phonography. Rev. ed. 1898. 19½cm Putnam \$2

Pitman, Sir Isaac. Isaac Pitman's complete phonographic instructor. Rev. ed. 1900. 16½cm Pitman \$1.50

— Shorthand dictionary; prepared by P. B. S. Peters. 1901. 15 x 8cm Powers \$1

Rose, R. F. Stenographers' aid; shorthand illustrations by J. L. Kapus. 1900. 20cm
Author, Chicago \$1

Torrey, Bates. Instruction in practical shorthand; a simple and progressive arrangement of the essentials of Graham phonography. 1902. 20cm Heath \$1.50 n

654 Telegraphs. Cables. Signals

Bright, Charles. Story of the Atlantic cable. 1903. 19½cm Appleton \$1 n

Houston, E. J., & Kennelly, A. E. Electric telegraphy. 1897. 17½cm (Elem. electro-tech. ser.) McGraw \$1

Duplex, quadruplex and multiplex methods described and illustrated, giving good idea of process of sending simultaneously several messages over same wire. Two chapters on submarine telegraphy. Map of world showing cable and some overland connections. Describes methods of laying and repairing cables. *Elec. world*, 29: 540.

Electric telephone. 2d ed. 1902. 17½cm (Elem. electro-tech. ser.) McGraw \$1

From elementary principles to the complete modern systems, including line work, switch boards and other apparatus. Chapter on radiophony, telephony with a beam of light as a carrier of wave impulses. *Elec. world*, 29: 428.

Lodge, Sir O. J. Signalling across space without wires; a description of the work of Hertz and his successors. 3d ed. 1900. 22cm

Van Nostrand \$2

Largely Royal institution lectures 1894. Account also of later applications of wave experiments to wireless telegraphy. Not severely technical. *Engin.* n. 46: 64.

Maver, William, jr. American telegraphy and encyclopedia of the telegraph. 5th ed. 1903. 24½cm Van Nostrand \$3.50

Clear and complete description of various systems and apparatus. Excellent for reference. For practical advanced students and engineers. Crocker.

Miller, K. B. American telephone practice. 3d ed. 1900. 26½cm Amer. elec. co. \$3

Clear and logical; entire field is carefully covered. *Engin.* n. 44: 84.

Williams, J. A. Manual of rural telephony. 1902. 18½cm Manual pub. co. \$1

Condenses instruction for organizing and financing telephone companies; intended to aid rural electrician. *Elec. world*, 41: 207.

Wittbecker, W. A. Domestic electrical work. 1902. 20cm Williams 50c

Appeared as articles in the *Metal worker*.

For amateur electrician. How to wire buildings for bells, alarms, annunciators and for gas lighting from batteries. Title.

655 Printing. Publishing. Copyright

De Vinne, T. L. Practice of typography. v. 1-3. 1900-02. 19½cm Century \$2 ea n

V. 1, Plain printing types; processes of type-making, point system, names, sizes, styles and prices of plain printing types. v. 2, Correct composition; a treatise on spelling, abbreviations, the compounding and division of words, the proper use of figures and numerals, italic and capital letters, notes, etc. with observations on punctuation and proof-reading. v. 3, Treatise on title-pages, with numerous illustrations in facsimile and some observations on the early and recent printing of books.

Pennell, Joseph. Modern illustration. 1895. 19½cm (Ex-libris ser.) Macmillan \$3.50 n

Putnam, G. H. Authors and their public in ancient times. 1894. 19½cm Putnam \$1.50

Principal works referred to, pref. p. 13-17.

Sketch of literary conditions and of the relations with the public of literary producers, from the earliest times to the invention of printing. Title.

Books and their makers during the middle ages. 2 v. 1896-97. 23½cm

Putnam \$2.50 ea

Bibliography, v. 1, pref. p. 17-27.

Study of conditions of production and distribution of literature from fall of Roman empire to close of 17th century. Title.

ed. Question of copyright. 2d ed. 1896. 20cm Putnam \$1.75

Comprising text of U. S. copyright law and a summary of copyright laws at present in force in the chief countries. Title.

& J. B. Authors and publishers. 7th ed. 1900. 21cm Putnam \$1.75

Contents: Publishing methods and arrangements, by G. H. P. The making of books, by J. B. P.

Manual of suggestions for beginners in literature, comprising a description of publishing methods and arrangements, together with general hints for authors. Title.

Rawlings, G. B. Story of books. 1901. 15½cm (Lib. of useful stories) Appleton 35c n

Outline history of making and preservation of books, manuscript and printed, to about 1550, with brief chapter on modern bookmaking. N. Y.

Thompson, J. S. Mechanism of the linotype: installation, operation and care. 1902. 20cm

Inland pr. co. \$1.50

Articles originally published in *Inland printer*, revised and amended to include latest improvements. Pref.

656 Transportation. Railroading

The American railway, its construction, development, management and appliances; with introd. by T. M. Cooley. 1897. 24½cm Scribner \$3

Reprinted from *Scribner's magazine*, 1888-89.

Papers by T. C. Clarke, John Bogart, M. N. Forney, E. P. Alexander, H. G. Prout, Horace Porter, Theodore Voorhees, Benjamin Norton, A. T. Hadley, T. L. James, C. F. Adams, B. B. Adams Jr.

Brooks, E. S. The American sailor. Rev. ed. 1899. 23cm Lothrop \$1.50

Published in 1888 with title *Story of the American sailor*.

Story of his development and deeds on merchant vessel and man of war to 1900.

Haines, H. S. American railway management. 1897. 19cm Wiley \$2.50

Addresses before American railway association, and miscellaneous addresses and papers. Title.

Knight, A. M. Modern seamanship. 1901. 24cm Van Nostrand \$6 n

Modern in method and in scope; a valuable manual for yachts and merchant vessels as well as for men of war. Pittsburgh.

Maginnis, A. J. The Atlantic ferry; its ships, men and working. 3d ed. 1900. 19½cm

Macmillan \$2

Relates to trans-Atlantic steam trade, giving account of respective lines, vessels and workings. Well illustrated. Important tables and chart show development of steamers. Van Nostrand, 21: 103.

Patterson, Howard. Patterson's illustrated nautical encyclopedia. Rev. ed. 1901. 26½cm

Marine rev. \$3

Rf A work of reference for naval revenue, coast survey, transport and light house services, and adapted for use of the naval reserve. Title.

Weyl, W. E. Passenger traffic of railways. 1901. 25cm (Penn. univ. ser. in pol. econ. & pub. law) Ginn \$1.50

Valuable as summary of statistics gathered from principal European countries and United States. R. of r. 25: 379.

657 Bookkeeping. Accounts

Arnold, H. L. Complete cost-keeper. 1900. 25cm Engin. m. \$5

Some original systems of shop cost-keeping or factory accounting, with an exposition of advantages of account keeping by means of cards instead of books. Title.

Dean, W. F. Bookkeeping for retail stores. 1900. 19cm Stern \$1

Practical treatise, with forms.

Gilbert, A. L. Modern business bookkeeping; text-book for schools and colleges. 1902. 25½cm (Bowen-Merrill ser.) Merrill \$1.50 n

Goodwin, J. H. Goodwin's improved bookkeeping and business manual. 10th ed. 1889. 26cm Baker & T. \$3

Keister, D. A. Keister's corporation accounting and auditing; with an introd. by H. C. White. 8th ed. 1901. 23½cm Burrows \$4 n

Practical treatise on higher accounting. Title.

Moore, J. H., & Miner, G. W. Accounting and business practice; for use in schools. 1902. 24cm Ginn \$1.40

Seavy, Manson. Practical business bookkeeping by double entry; for schools and for self-instruction. 1893. 26x20½cm

Heath \$1.40 n

Soulé, George. Soulé's new science and practice of accounts. 7th ed. 1903. 27cm

Caspar \$4 n

Exposition of double entry and single entry bookkeeping with the most approved forms of merchandising, com-

mission and other lines of business. Joint stock company and corporation bookkeeping and expert work in corporation accounting and limited partnership companies are special features. Title.

658 Business manuals. Methods. Tables

Cromwell, J. H. American business woman. 1900. 22cm Putnam \$2

An endeavor to teach women in a serious but readable way how to handle bank accounts and to deal with real and personal property, with cautious advice as to general principles of investment. Nation, 71: 191 (B)

Hooper, Frederick, & Graham, James. Introduction to modern business methods: the home trade. 1901. 22cm (Mod. man. of commerce) Macmillan \$1

Guide to operations incidental to the trade of the United Kingdom, with customary documents and correspondence. A reference book for business men and a text-book for commercial students. Title.

Palmer, O. R. Type-writing and business correspondence. Rev. ed. 1900. 25½cm

Lippincott \$1

Business letters based on actual correspondence, giving commercial forms and expressions; brief rules for punctuation; directions for proper fingering of keys; examples of law forms, statements, specifications, etc. reproduced from type-written work. Pref.

659 Advertising

Fowler, N. C. Building business. 1893. 23x18cm Publicity pub. co. \$1.50

Comprehensive and practical book about advertising. Buffalo.

660 Chemical technology

Blount, Bertram. Practical electro-chemistry. 1901. 23cm Macmillan \$3.25 n

Discusses economic aspects of every branch of electro-chemistry. Masterly discussion of economics of electricity. Nation, 73: 173.

— & **Bloxam, A. G.** Chemistry for engineers and manufacturers. 2v. 1900. 22cm (Griffin's sci. text-bks.)

Lippincott v. 1, \$3; v. 2, \$4

Bibliography, 2: 438-41.

V. 1 includes chemistry of materials of construction, of fuels, waters, lubricants, and the metallurgy of iron, copper, lead, zinc, tin, mercury, silver, gold, etc. Engin. n. 35: 392.

V. 2 treats briefly chemistry of manufacture of sulphuric acid, alkali, artificial manure, lime and cement, glass, sugar and starch, oils, etc. also of destructive distillation, petroleum, clay industries, brewing and distilling, textiles and bleaching. Handy manual for general reader. Engin. n. 36: 424.

Cremer, J. H., & Bicknell, G. A. Chemical and metallurgical hand book; containing tables, formulas and information for the use of chemists, metallurgists and mining engineers. 3d ed. 1903. 16½cm Authors, Cleveland \$3.50 n

Meldola, Raphael. Coal and what we get from it. 2d ed. 1897. 17½cm (Romance of sci.) E. & J. B. Young \$1

Origin of coal; history and manufacture of coal gas; history and description of substances obtained from coal tar. *Knowl.* 14: 172.

Sadtler, S. P. Hand-book of industrial organic chemistry. 3d ed. 1900. 24½cm Lippincott \$5 n

Bibliography at end of each chapter.

Describes raw materials and process of manufacture and products of several industries, and gives analytic tests and methods, bibliography and statistics of important industries. *Engin. n.* 36: 352.

Thorp, F. H. Outlines of industrial chemistry; a text-book for students. 1898. 23cm Macmillan \$3.50 n

Brief bibliography, pref. p. 19.

Treats subjects under "Inorganic industries" and "Organic industries." Descriptions of processes, concise, clear and interesting. Frequent reference to American practice an important feature. W. A. Noyes in *Science*, 32: 150.

Thorpe, T. E. Dictionary of applied chemistry. 3 v. 1898-1900. 23½cm Longmans \$50

Rf Essentially a dictionary of chemistry in its application to arts and manufactures. Complementary to Watt's *Dictionary of chemistry*, general plan and arrangement similar. Pref.

Wagner, Rudolf von. Manual of chemical technology; tr. and ed. by Sir William Crookes. 1900. 26½cm Appleton \$7.50

662 Pyrotechnics, explosives, etc.

Poole, Herman. Calorific power of fuels. 2d ed. 1900. 23½cm Wiley \$3

Authorities consulted, pref. p. 15-17.

Based on translation of *Pouvoir calorifique des combustibles* by M. Scheurer-Kestner, with changes necessary to adapt it to American methods and data, omitting many portions and adding new matter. *Engin. n.* 39: 360.

663 Beverages

Matthews, C. G. Manual of alcoholic fermentation and the allied industries. 1902. 19cm Longmans \$2.60 n

665 Oils. Gases

Hornby, John. Text-book of gas manufacture. 1902. 18cm (Tech. handbks.) Macmillan \$1.50

Concise little book setting forth the chief points in gas manufacture in a manner that students may readily grasp. F. H. Thorp in *Science*, 26: 815.

666 Ceramics. Glass

Bolas, Thomas. Glass blowing and working. 1898. 18½cm Lane \$1 n

Bibliography, p. 207-9.

Chapters on glass work in miniature, tools and appliances, methods of working and blowing glass, coloring and modifying materials, making decorative articles, etc. Practical guide. Illustrated with outline drawings. N. Y.

Davis, C. T. Practical treatise on the manufacture of bricks, tiles, terra-cotta, etc. 2d ed. 1889. 24cm Baird \$5

Meade, R. K. Chemical and physical examination of Portland cement. 1901. 15 x 12½cm Chemical pub. co. \$1

Describes standard methods in detail. Intended as guide for inexperienced chemists and engineers. Pittsburgh.

Powell, H. J. Principles of glass-making; with treatises on crown and sheet glass, by Henry Chance, and plate glass, by H. G. Harris. 1883. 18cm (Tech. handbks.) Macmillan \$1

Randau, Paul. Enamels and enamelling; tr. by Charles Salter. 1900. 22cm Van Nostrand \$4 n

Introduction to preparation and application of all kinds of enamels for technical and artistic purposes. Title.

667 Bleaching, dyeing, etc.

Brannt, W. T. Varnishes, lacquers, printing inks and sealing-waxes. 1893. 19cm Baird \$3

Includes description of properties of raw materials used and simple methods for testing them. Appendix gives receipts for putties, stains for wood, bone and ivory, etc. Pref.

Hurst, G. H. Garment dyeing and cleaning. 2d ed. 1901. 20cm Lippincott \$1.75 n

Practical work by English chemist. Treats also of cleaning garments, bleaching and finishing fabrics, scouring and dyeing skin rugs and mats, feathers and gloves, and straw bleaching and dyeing. Fully illustrated. L. & I.

Painters' colours, oils and varnishes; a practical manual. 3d ed. 1901. 20cm Lippincott \$3.50 n

Lehner, Sigmund. Ink manufacture; tr. from the German by W. T. Brannt. 1892. 19cm Baird \$2

Deals with raw materials, chemical constitution, various kinds of ink, preserving agents, extracts and powders, etc. Most of the receipts have been practically tested. Pref.

Eng. ed. 1902 tr. by Arthur Morris and Herbert Robson, pub. by Scott, Greenwood & co. 5 n